

- Q.26 Describe the process of calibrating a rotameter.
- Q.27 What is a thermistor, and where is it commonly used in process industries?
- Q.28 How is viscosity typically measured in fluids?
- Q.29 Describe a method for measuring conductivity in food samples.
- Q.30 Describe a common instrument used for measuring humidity.
- Q.31 What does pH value indicate in a solution?
- Q.32 Define automatic process control and explain its importance in industrial settings.
- Q.33 What does TSS stand for, and what does it measure?
- Q.34 Why is the accuracy of a thermometer so important in the food processing industry?
- Q.35 What are static characteristics of an instrument?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Describe the classification of instruments based on their function, such as measurement and control.
- Q.37 What is a venture meter? Explain its working with the help of diagram.
- Q.38 Explain the working principles of manometers and Bourdon gauges for measuring pressure.

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Roll No.

5th Sem / Food Tech.

Subject:- Instrumentation and Process Control

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Why are instruments crucial in process industries?
- To increase production costs
 - To decrease productivity
 - To monitor and control various parameters
 - To introduce complexity
- Q.2 Which classification distinguishes instruments based on their method of measurement?
- Analog vs digital
 - Static vs dynamic
 - Direct vs indirect
 - Electrical vs. mechanical
- Q.3 Which instrument is commonly used to measure pressure in pipelines and vessels?
- Manometer
 - Bourdon gauge
 - Pitot tube
 - Venturi meter
- Q.4 What is automatic process control?
- Manual adjustment of process variables
 - Continuous monitoring and adjustment of process variables
 - Irregular adjustment of process parameters
 - Disregard for process variations

- Q.5 What is the primary function of a thermometer?
 a) Measure pressure b) Measure flow rate
 c) Measure temperature d) Measure humidity
- Q.6 In which of the following temperature measurement methods is contact with the object not required?
 a) Thermometer b) Thermocouple
 c) Thermistor d) pyrometer
- Q.7 Which instrument is commonly used to measure the viscosity of fluids?
 a) pH meter b) Viscometer
 c) Conductivity meter d) Hygrometer
- Q.8 What does conductivity measure in a solution?
 a) pH level
 b) Humidity
 c) Ability to conduct electricity
 d) Viscosity
- Q.9 What unit is typically used to express humidity levels?
 a) Pascal (Pa) b) Kelvin (K)
 c) Percentage (%) d) Siemens (S)
- Q.10 A solution with a pH value below 7 is considered:
 a) Neutral b) Acidic
 c) Basic d) Saline

SECTION-B

- Note:** Objective type questions. All questions are compulsory. (10x1=10)
- Q.11 Thermocouples are sensitive to changes in temperature and provide accurate measurements. (True/False)
- Q.12 Thermometers are suitable for measuring high temperatures in industrial processes. (True/False)

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- Q.13 Manometers are used to measure pressure by comparing it to the pressure of a column of liquid. (True/False)
- Q.14 Rotameters measure flow rate based on the pressure difference between two points. (True/False)
- Q.15 Pitot tubes are suitable for measuring flow rates in closed conduits. (True/False)
- Q.16 Conductivity meters are used to measure the concentration of dissolved solids in a solution. (True/False)
- Q.17 A _____ measures temperature using the expansion or contraction of a liquid, such as mercury or alcohol. (Thermometer/ Thermocouple)
- Q.18 A _____ is a type of pressure gauge that measures pressure by the deflection of a flattened tube. (Bourdon gauge/ Pyrometer)
- Q.19 The _____ meter is a type of flow meter that measures the rate of fluid flow by forcing the fluid through a small opening. (Orifice/ Venturi)
- Q.20 _____ is the ability of a solution to conduct electrical current, which is influenced by the concentration of ions present. (Conductivity)

SECTION-C

- Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)
- Q.21 Discuss the importance of instruments in process industries?
- Q.22 How does a pyrometer measure temperature?
- Q.23 Describe the working principle of an orifice meter.
- Q.24 How is viscosity important in food processing?
- Q.25 What are the key differences between a thermometer and a thermocouple?

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